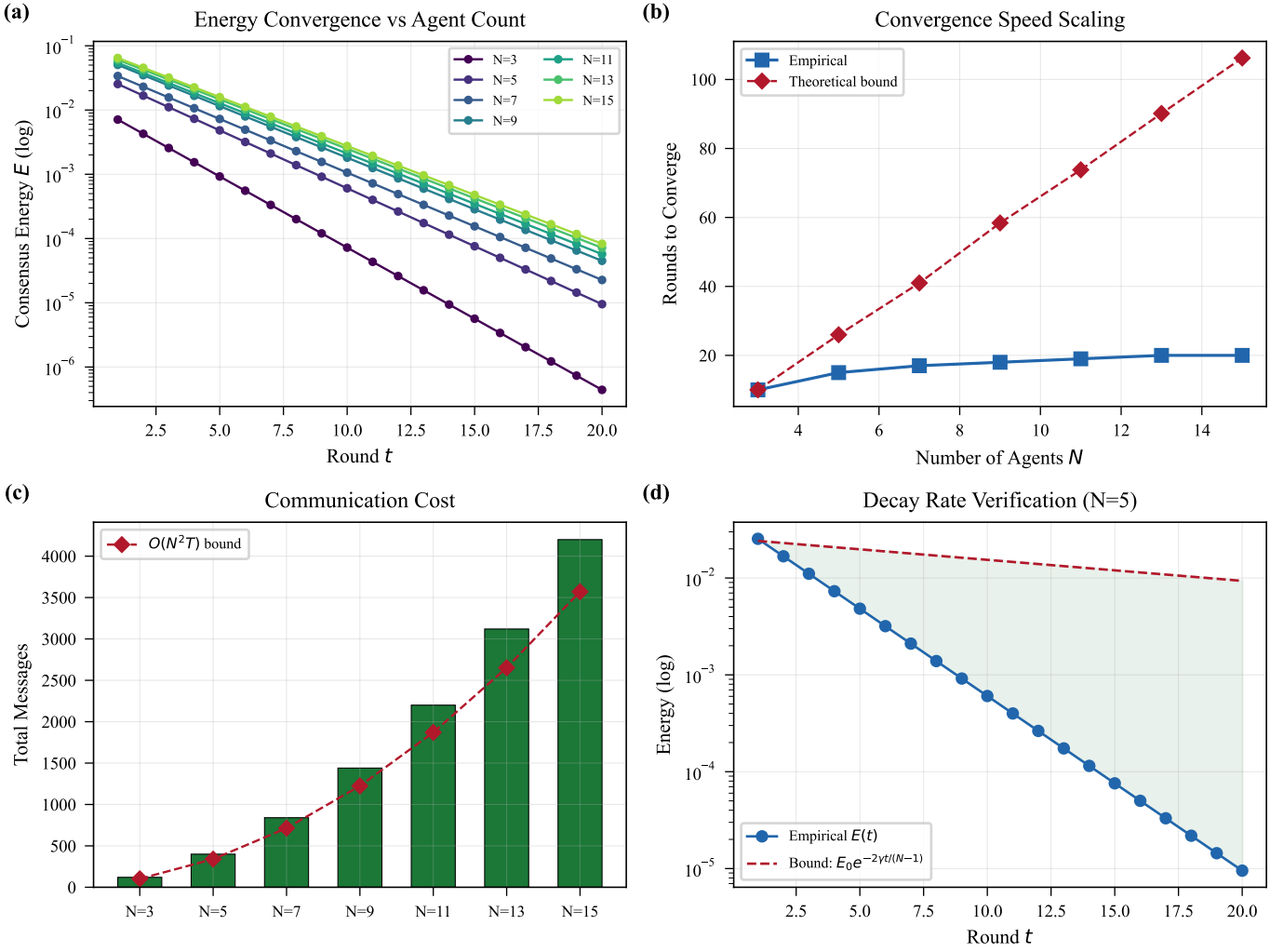
(a)
Energy Convergence vs Agent Count
Consensus Energy E (log)
Round t
N=3
N=5
N=7
N=9
N=11
N=13
N=15
(b)
Convergence Speed Scaling
Rounds to Converge
Number of Agents N
Empirical
Theoretical bound
(c)
Communication Cost
Total Messages
$O(N^2T)$ bound
(d)
Decay Rate Verification (N=5)
Energy (log)
Round t
Empirical $E(t)$
Bound: $E_0e^{-2\gamma t/(N-1)}$